The Gravity Equation and Gains from Trade

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Outline

1 **Gravity: Facts**

2 Micro-foundations

3 Structural Gravity

4 Gains from Trade (ACR)

5 Taking ACR to the Data

Gravity Equations

- $X_{ni} = G \underbrace{S_i}_{\text{cap. of exporter } i} \times \underbrace{M_n}_{\text{size of market } n} \times \underbrace{\phi_{ni}}_{\text{bilateral distance and more}}$
- Multiplicative separable: $\ln X_{ni} = \ln G + \ln S_i + \ln M_n + \ln \phi_{ni}$
- Typically approximate S_i and M_n with a country's GDP
- Empirical regularity first found by Tinbergen (1962)
- Term Gravity Equation $\sim$ Analogy with force between two bodies in physics, which is proportional to the product of their masses and inversely proportional to their distance

Gravity Equations

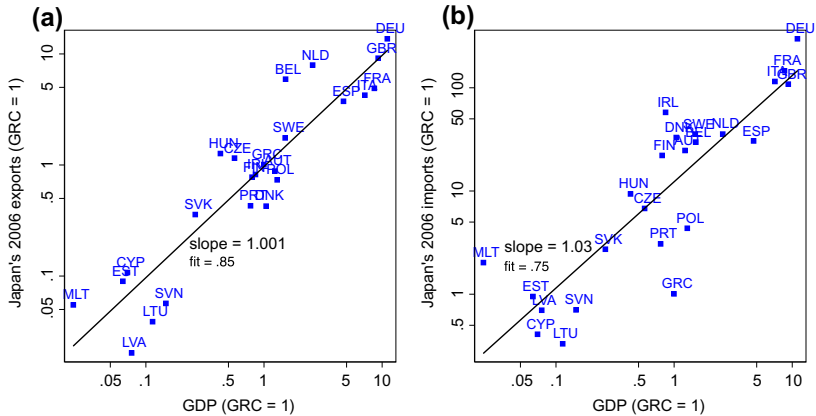


Figure 3.1 Trade is Proportional to Size; (a) Japan's Exports to EU, 2006; (b) Japan's Imports from EU, 2006. GRC: Greece

Gravity Equations

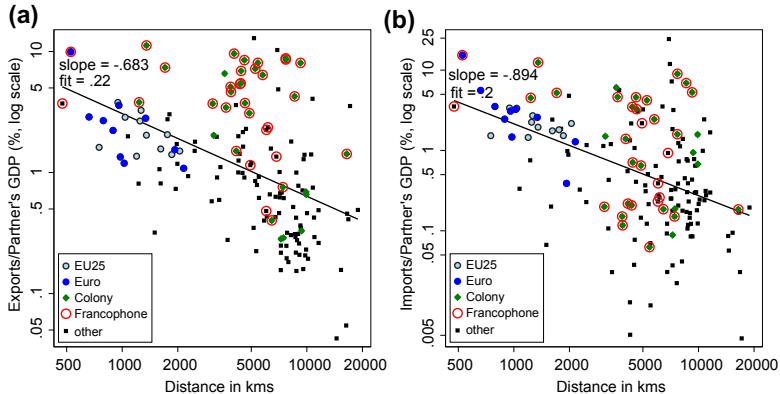


Figure 3.2 Trade is Inversely Proportional to Distance; (a) France's Exports (2006); (b) France's Imports (2006)

Gravity Equations

- Applications: infer trade flow effects of
  - Customs unions
  - Exchange-rate mechanisms
  - Ethnic ties
  - Linguistic identity
  - International borders
  - World Trade Organization (WTO) membership

What Do Gravity Regressions Find? A Meta-Analysis

Head and Mayer (2014): 2,508 coefficients from 159 papers. "Structural" = papers using country fixed effects (or ratio-type estimators).

	All gravity		Structural gravity	
	median	mean	median	mean
Origin GDP	0.97	0.98	0.86	0.74
Destination GDP	0.85	0.84	0.67	0.58
Distance	-0.89	-0.93	-1.14	-1.10
Contiguity	0.49	0.53	0.52	0.66
Common language	0.49	0.54	0.33	0.39
Colonial link	0.91	0.92	0.84	0.75
RTA/FTA	0.47	0.59	0.28	0.36
Common currency	0.87	0.79	0.98	0.86
Home (border)	1.93	1.96	1.55	1.90

- GDP elasticities close to 1; distance elasticity close to -1
- Dummies are large: $e^{0.5} - 1 \approx 65\%$ more trade for a shared border or language; $e^{1.55} \approx 4.7$ times more trade within than across national borders

The (Puzzling) Persistence of Distance

- Disdier and Head (2008): meta-analysis of 1,467 distance elasticities from 103 papers
  - Mean estimate about -0.9; the distance effect rose around the middle of the century and has stayed high since
  - Despite large declines in freight and communication costs: "death of distance" is not in the data

- Why is the elasticity so stable and so close to -1?
  - In the micro-founded models below, $\partial \ln X_{ni} / \partial \ln dist_{ni} = \varepsilon \times \rho$: (trade elasticity) $\times$ (elasticity of trade costs w.r.t. distance); nothing pins this product at -1
  - Chaney (2018): if the firm size distribution is Zipf and firms expand their reach to distant markets through networks of contacts, the aggregate distance elasticity is ≈ -1 regardless of physical trade costs

- Natural experiments: Feyrer (2019) uses the 1967–75 closure of the Suez Canal, which changed sea distances for some country pairs but not others, to identify distance effects (and trade's effect on income)

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Gravity Equations: Micro-foundations

- Probably the simplest starting point is the **Armington** (1969) model
- Goods differentiated by country of origin price p_i, exogenous Y_i.
- $U_n = \left[\sum_{i=1}^{N} q_{ni}^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$ and $P_n = \left[\sum_{i=1}^{N} \left(p_i \tau_{ni}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$
- Consumer purchase $X_{ni} = \left(\frac{p_i \tau_{ni}}{P_n}\right)^{1-\sigma} Y_n$, where $Y_n = \sum_{i=1}^{N} X_{ni}$ total expenditure in country n
- Anderson and van Wincoop (2003) assumes that balanced trade:

$$\underbrace{Y_i = \sum_{l=1}^{N} X_{li}}_{\text{balanced trade}} = \sum_{l=1}^{N} \left(\frac{p_i \tau_{li}}{P_l}\right)^{1-\sigma} Y_l \rightarrow p_i^{1-\sigma} = \left(\frac{Y_i}{\sum_l \left(\frac{\tau_{li}}{P_l}\right)^{1-\sigma} Y_l}\right)$$

- Substitute back to $X_{ni} = Y_n \frac{\left(\frac{\tau_{ni}}{P_n}\right)^{1-\sigma} Y_i}{\sum_l \left(\frac{\tau_{li}}{P_l}\right)^{1-\sigma} Y_l} \equiv \frac{Y_i Y_n}{Y^W} \left(\frac{\tau_{ni}}{P_n \Pi_i}\right)^{1-\sigma}$

Gravity Equations: Micro-foundations

- We define

$$\Pi_i^{1-\sigma} = \sum_l \left(\frac{\tau_{li}}{P_l}\right)^{1-\sigma} \frac{Y_l}{Y^W}$$

- In turn, we can also write the price index as

$$P_n^{1-\sigma} = \sum_i \left(\frac{\tau_{ni}}{\Pi_i}\right)^{1-\sigma} \frac{Y_i}{Y^W}$$

- The equilibrium can be defined by Π_i and P_n – when $\tau_{in} = \tau_{ni}$, these two multilateral resistance terms overlap.
- This micro-founds the "structural gravity equation", where Π_i is sometimes called "market access".

Anderson and van Wincoop (2003)

$$X_{ni} = \frac{Y_i Y_n}{Y_w} \left(\frac{\tau_{ni}}{P_i P_n} \right)^{1-\sigma} \quad \text{(with symmetric } \tau_{ni} = \tau_{in}, \Pi_i = P_i\text{)}$$

- Size matters: larger (richer) countries import more and export more.
- Trade barriers matter: bilateral trade barriers (τ_{ni} large) reduce trade.
- For given bilateral trade barriers τ_{ni}:
  - Country n imports more from i the larger is P_n
    - less competition
  - Country i exports more to n the larger is P_i
    - market access
- Key ingredient: price indices depend on relative sizes and relative trade barriers.

Anderson and van Wincoop (2003): Empirics

- $X_{ni} = \frac{Y_i Y_n}{Y_w} \left(\frac{\tau_{ni}}{P_i P_n} \right)^{1-\sigma}$
- $\tau_{ni} = b_{ni}\, dist_{ni}^{\rho}$, where $b_{ni} = 1 + b$ if n and i are in different countries, 1 otherwise

$$\Rightarrow \ln X_{ni} = k + \ln Y_i + \ln Y_n + (1-\sigma) \ln (1+b) (1-\delta_{ni}) + (1-\sigma)\, \rho \ln dist_{ni} - (1-\sigma) \ln P_i - (1-\sigma) \ln P_n$$

$$\Rightarrow \ln X_{ni} = k + \ln Y_i + \ln Y_n + \underbrace{a_1 (1-\delta_{ni})}_{\text{border}} + \underbrace{a_2 \ln dist_{ni}}_{\text{distance}} - (1-\sigma) \ln P_i - (1-\sigma) \ln P_n$$

- Regressions without P_i and P_n suffer from omitted variable bias: P_i, P_n are correlated with $dist_{ni}$ and the border dummy. The paper shows this insight resolves the "border puzzle" (McCallum, 1995).

The Border Puzzle: McCallum (1995) vs. Anderson–van Wincoop (2003)

- McCallum (1995): 1988 data on Canadian provinces and U.S. states, "naive" gravity with GDPs and distance
  - Inter-provincial trade is a factor of **22** larger than province–state trade, controlling for size and distance
- Why is this "puzzling"? A tariff-free, culturally similar border should not cut trade by 95%
- AvW (2003): the border raises *relative* trade costs, and its effect on a pair depends on multilateral resistance
  - Canada is small: the border raises Canada's multilateral resistance a lot $\Rightarrow$ large increase in intra-Canadian trade (factor ≈ 6)
  - The U.S. is large: intra-U.S. trade rises by only $\approx 25\%$
  - Borders reduce U.S.–Canada trade by 44% and trade among other industrialized countries by 29% (benchmark $\sigma = 5$)
- Lesson: the border coefficient in a gravity regression is not a sufficient statistic for the effect of the border on trade flows; one has to solve for the general-equilibrium change in P_i, P_n

Gravity Equations: Micro-foundations

- Because of the strong regularity (and good fit) of gravity equations, the international trade field sees it as a plus if a model implies a gravity equation.

- We saw that the Eaton and Kortum (2002) does imply one:

$$X_{ni} = \pi_{ni} X_n = \frac{T_i (c_i \tau_{ni})^{-\theta}}{\Phi_n} X_n$$

  where $\Phi_n = \sum_{i'} T_{i'} (c_{i'} \tau_{ni'})^{-\theta}$ and P_n is proportional to $\Phi_n^{-1/\theta}$.

- Grounding the gravity equations into general equilibrium models helps understand how to conduct counterfactuals (i.e. Hat Algebra).

Gravity Equation: Micro-foundations

- Multiple country CES-Krugman (1980), Country i produces N_i goods and $U_n = \sum_{i=1}^{C} \sum_{k=1}^{N_i} (c_{ni,k})^{\frac{\sigma-1}{\sigma}}$
- All products exported by country i sell for p_i net of any transport cost and $p_{ni} = p_i \tau_{ni}$ with $\tau_{ni} \geq 1$ and $\tau_{ii} = 1$
- Homogeneous firms: $c_{ni,k} = c_{ni} \; \forall k = 1, ..., N_i$
- Consumer in country n maximizes $U_n = \sum_{i=1}^{C} N_i \left(c_{ni}\right)^{\frac{\sigma-1}{\sigma}}$ subject to budget constraint $\sum_{i=1}^{C} N_i p_{ni} c_{ni} = X_n = w_n L_n$
- Consumer and firm optimization has $p_{ni} = w_i \tau_{ni} \frac{\sigma}{\sigma-1}$ and $q_{ni} = c_{ni} = \frac{X_n}{P_n} \left(\frac{p_{ni}}{P_n} \right)^{-\sigma}$.

Gravity Equation: Micro-foundations

- We derive the trade flow: $X_{ni} = N_i p_{ni} q_{ni} = N_i \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} (w_i \tau_{ni})^{1-\sigma} \frac{w_n L_n}{P_n^{1-\sigma}}$
- $X_n = \sum_i X_{ni} = \sum_i N_i \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} (w_i \tau_{ni})^{1-\sigma} \frac{w_n L_n}{P_n^{1-\sigma}}$
- $\pi_{ni} = \frac{X_{ni}}{X_n} = \frac{\left(N_i w_i^{1-\sigma}\right) \tau_{ni}^{1-\sigma}}{\sum_\ell \left(N_\ell w_\ell^{1-\sigma}\right) \tau_{n\ell}^{1-\sigma}}$
- With free entry (fixed cost F units of labor), $N_i = L_i/(\sigma F)$: the number of varieties is proportional to country size
- Same functional form as Armington with $p_i^{1-\sigma}$ replaced by $N_i w_i^{1-\sigma}$; trade elasticity is again $1 - \sigma$

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Structural Gravity: Head and Mayer (2014)

- Structural Gravity: useful to have a general gravity model in which bilateral trade flows obey (Next slide we show how we can obtain it.)

$$X_{ni} = \frac{Y_i}{\Omega_i}\frac{X_n}{\Phi_n}\phi_{ni}$$

- $Y_i = \sum_n X_{ni}$ - value of production
- $X_n = \sum_i X_{ni}$ - value of importer's expenditure
- Ω_i and Φ_n are called "multilateral resistance", where $\Phi_n = \sum_\ell \frac{\phi_{n\ell} Y_\ell}{\Omega_\ell}$ and $\Omega_i = \sum_\ell \frac{\phi_{\ell i} X_\ell}{\Phi_\ell}$
- Large set of models fit this definition: Armington, Krugman, Eaton and Kortum, Melitz-Chaney
- Ω_i and Φ_n depend on $\phi_{ni} \Rightarrow$ allows for a more comprehensive calculation of the impact of trade cost changes.

Structural Gravity: Head and Mayer (2014)

- Spatial allocation of expenditure for the importer

$$X_{ni} = \pi_{ni} X_n \text{ (identity)}$$

- Assumption: π_{ni} can be expressed multiplicatively as

$$\pi_{ni} = \frac{S_i \phi_{ni}}{\Phi_n} \text{ where } \Phi_n = \sum_{\ell} S_\ell \phi_{n\ell}$$

- Sum of i's exports to all destinations equals value of i's production

$$Y_i = \sum_n X_{ni} = S_i \underbrace{\sum_n \frac{\phi_{ni} X_n}{\Phi_n}}_{\Omega_i} \Rightarrow S_i = \frac{Y_i}{\Omega_i}$$

- We can also write Φ_n in terms of Ω_i using $\Rightarrow \Phi_n = \sum_i S_i \phi_{ni} = \sum_i \frac{\phi_{ni} Y_i}{\Omega_i}$

- Substituting into π_{ni}, we have $X_{ni} = \frac{S_i \phi_{ni}}{\Phi_n} X_n \equiv \frac{Y_i}{\Omega_i} \frac{X_n}{\Phi_n} \phi_{ni}$

Structural Gravity: Summary

The above relationship covers all workhorse models we have learned so far

- Armington: $S_i = p_i^{1-\sigma}$, $\phi_{ni} = \tau_{ni}^{1-\sigma}$
- Krugman + CES: $S_i = N_i w_i^{1-\sigma}$, $\phi_{ni} = \tau_{ni}^{1-\sigma}$
- Simple Eaton-Kortum: $S_i = T_i w_i^{-\theta}$, $\phi_{ni} = \tau_{ni}^{-\theta}$
- Melitz-Chaney (Pareto shape θ, lower bound b_i, fixed cost f_{ni} paid in destination labor): $S_i = N_i^E b_i^\theta w_i^{-\theta}$, $\phi_{ni} = \tau_{ni}^{-\theta} f_{ni}^{-\left(\frac{\theta}{\sigma-1}-1\right)}$

Conditional on the fact that we observe Y_i and X_n, we now proceed to define the iterative procedure to estimate the model.

Iterative Structural Estimation: Armington Revisited

- $X_{ni} = \frac{Y_i X_n}{Y_w} \left(\frac{\tau_{ni}}{P_i P_n} \right)^{1-\sigma}$
- let $\tau_{ni} = b_{ni}\, dist_{ni}^{\rho}$
  - Where $b_{ni} = \begin{cases} 1 \text{ if } n \text{ and } i \text{ are in the same country} \\ 1 + b \text{ otherwise} \end{cases}$

Iterative Structural Estimation: Armington Revisited

$$\ln X_{ni} = k + \ln Y_i + \ln X_n + (1-\sigma)\ln(1+b)(1-\delta_{ni}) + (1-\sigma)\rho \ln dist_{ni} - (1-\sigma)\ln P_i - (1-\sigma)\ln P_n$$

$$\ln X_{ni} = k + \ln Y_i + \ln X_n + a_1(1-\delta_{ni}) + a_2 \ln dist_{ni} - (1-\sigma)\ln P_i - (1-\sigma)\ln P_n$$

$$a_1 = (1-\sigma)\ln(1+b)$$
$$a_2 = (1-\sigma)\rho$$
$$\delta_{ni} = \begin{cases} 1 \text{ if } n \text{ and } i \text{ are in the same country} \\ 0 \text{ otherwise} \end{cases}$$

Iterative Structural Estimation: Armington Revisited

- Assume $\Omega_i = 1$ and $\Phi_n = 1$
  - $P_i^{1-\sigma} = 1$ and $P_n^{1-\sigma} = 1$
- Estimate a_1 and a_2 by OLS

$$\ln X_{ni} - \ln Y_i - \ln X_n = k + a_1 (1 - \delta_{ni}) + a_2 \ln dist_{ni} + u_{ni}$$

- Use $P_n^{1-\sigma} = \sum_i \frac{Y_i}{Y_w} \left(\frac{\tau_{ni}}{P_i} \right)^{1-\sigma}$ and $P_i^{1-\sigma} = \sum_n \frac{X_n}{Y_w} \left(\frac{\tau_{ni}}{P_n} \right)^{1-\sigma}$, with $\tau_{ni}^{1-\sigma} = e^{a_1(1-\delta_{ni})} dist_{ni}^{a_2}$
- In order to solve for new guesses of Ω_i and Φ_n ($P_i^{1-\sigma}$ and $P_n^{1-\sigma}$)
- Re-estimate with $P_i^{1-\sigma}$ and $P_n^{1-\sigma}$; repeat until convergence.
- Similar procedure for any of the other micro-founded models.

Estimation in Practice I: Fixed Effects

- Iterating on the multilateral resistance terms is not necessary if we only need a_1, a_2: absorb them with fixed effects

$$\ln X_{ni} = \underbrace{\ln \frac{Y_i}{\Omega_i}}_{\text{exporter FE}} + \underbrace{\ln \frac{X_n}{\Phi_n}}_{\text{importer FE}} + \ln \phi_{ni} + u_{ni}$$

- Consistent for ϕ_{ni} parameters under any model in the structural gravity class
- Cannot identify the effect of country-level variables (e.g., a country's tariff on all partners, landlockedness); these are absorbed by the FE

- Panel data: exporter-year and importer-year FE (multilateral resistance moves over time) plus **pair FE**
  - Pair FE control for unobserved, time-invariant bilateral frictions correlated with policy (endogeneity of trade agreements)
  - Baier and Bergstrand (2007): with pair FE, an FTA roughly doubles members' bilateral trade after 10 years

Estimation in Practice II: The Log of Gravity

- The model is a statement about the conditional mean: $X_{ni} = \exp(\mathbf{z}'_{ni}\beta)\,\eta_{ni}$ with $E[\eta_{ni} \mid \mathbf{z}_{ni}] = 1$
- Log-linearizing: $\ln X_{ni} = \mathbf{z}'_{ni}\beta + \ln \eta_{ni}$
  - Jensen's inequality: $E[\ln \eta_{ni} \mid \mathbf{z}_{ni}]$ depends on the variance of η_{ni}
  - If the error is heteroskedastic (variance depends on $\mathbf{z}_{ni}$), OLS on logs is **inconsistent**, even with FE
  - Zero trade flows are dropped (or "fixed" with $\ln(1 + X_{ni})$, which has no justification)
- Santos Silva and Tenreyro (2006): Poisson pseudo-maximum likelihood (PPML)

$$\sum_{n,i} \left[X_{ni} - \exp(\mathbf{z}'_{ni}\hat{\beta}) \right] \mathbf{z}_{ni} = 0$$

  - Consistent as long as the conditional mean is correctly specified; data need not be Poisson or integers
  - Estimated in levels, so zeros are included
  - In their application, PPML gives smaller distance elasticities and a smaller role for colonial ties than OLS

PPML with Fixed Effects Is Structural: Fally (2015)

- PPML with exporter FE e_i and importer FE m_n: $E[X_{ni}] = \exp(e_i + m_n + \beta' \mathbf{z}_{ni})$
- First-order conditions with respect to each fixed effect:

$$\sum_n X_{ni} = \sum_n \hat{X}_{ni} = Y_i, \qquad \sum_i X_{ni} = \sum_i \hat{X}_{ni} = X_n$$

- The fitted values exactly match each country's output and expenditure, which are the adding-up constraints that define Ω_i and Φ_n
- Hence $\exp(\hat{e}_i) = Y_i/\hat{\Omega}_i$ and $\exp(\hat{m}_n) = X_n/\hat{\Phi}_n$ (up to a normalization): the estimated FE *are* the multilateral resistance terms
  - No iteration needed; OLS FE do not have this property
- Implementation with many FE: `ppmlhdfe` in Stata (Correia, Guimarães and Zylkin, 2020), `fixest::fepois` in R, `pyfixest` in Python
- Panel with three-way FE: PPML is consistent, but the incidental parameter problem produces an asymptotic bias in point estimates and downward-biased standard errors; Weidner and Zylkin (2021) provide corrections

Zeros and the Extensive Margin

- About half of country pairs record zero trade in a given year at the aggregate level; far more at the product level
- CES-Armington, EK and Krugman predict strictly positive flows for any finite τ_{ni}
- Helpman, Melitz and Rubinstein (2008): Melitz with fixed export costs and bounded productivity
  - Only pairs where the most productive firm can cover f_{ni} trade $\Rightarrow$ selection into positive flows
  - Two-step estimation: probit for $\Pr(X_{ni} > 0)$, then gravity with (i) a Heckman correction and (ii) a control for the fraction of exporting firms (extensive margin)
  - Exclusion restriction: variables shifting fixed but not variable costs (regulation costs of firm entry, common religion)
  - Ignoring selection and firm heterogeneity biases standard gravity estimates

Zeros and the Extensive Margin: Distorted Gravity

- Chaney (2008), "Distorted gravity": Melitz with Pareto productivity (shape $\theta > \sigma - 1$) and fixed export costs,

$$\frac{\partial \ln X_{ni}}{\partial \ln \tau_{ni}} = \underbrace{-(\sigma - 1)}_{\text{intensive}} \underbrace{- [\theta - (\sigma - 1)]}_{\text{extensive}} = -\theta$$

  - Aggregate trade elasticity does not depend on σ: a higher σ raises the intensive margin response but dampens the extensive margin one

- Implications for estimation and interpretation
  - The coefficient on $\ln \tau_{ni}$ in a gravity regression identifies the Pareto shape θ, not the elasticity of substitution σ
  - Fixed export costs enter the bilateral term: $\phi_{ni} = \tau_{ni}^{-\theta} f_{ni}^{-\left(\frac{\theta}{\sigma-1}-1\right)}$, so "distance" and "language" coefficients mix variable- and fixed-cost effects
  - With firm-level data, the two margins can be measured separately (number of exporters vs. average sales per exporter)

Gravity as a Policy Evaluation Tool: Selected Findings

- Trade agreements: Baier and Bergstrand (2007) $\Rightarrow$ FTAs are endogenous; cross-sectional estimates substantially understate their effect, while panel estimates with pair FE imply roughly a doubling of members' trade after 10 years
- WTO/GATT membership: Rose (2004) finds no robust effect; Subramanian and Wei (2007) find large but uneven effects (industrial countries, members that liberalized)
- Colonial ties: Head, Mayer and Ries (2010) show trade between former colonies and their metropoles erodes gradually over decades after independence
- Practical guides: Head and Mayer (2014, "cookbook"); Yotov, Piermartini, Monteiro and Larch (2016), *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model* (WTO/UNCTAD)
- Recommended specification today: PPML, exporter-time and importer-time FE, pair FE, include intra-national flows X_{nn} when available (identifies border effects and non-discriminatory policies)

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Arkolakis, Costinot, Rodriguez-Clare (AER 2012)

- A lot of progress has been done uncovering the patterns of international trade at the micro data (e.g., Bernard, Jensen, Redding and Schott (2007, 2012)).
- We also had a lot of progress understanding these new patterns with models of international trade (Melitz (2003), Chaney (2008), Bernard Redding and Schott (2007 and 2011)).
- Incorporating love of variety, heterogeneous firms, multi-country and multi-sector general equilibrium analysis of trade policy, etc...
- Arkolakis, Costinot and Rodriguez Clare have a provoking argument
- All these models yield the same formulas for gains from trade!

Arkolakis, Costinot, Rodriguez-Clare (AER 2012)

- They set up a wide class of models of international trade
  - The Armington model, the Krugman (1980) model, the Eaton and Kortum (2002) model and the Melitz (2003) model are all particular cases of that class
- They show that all models within that class have gains from trade given by the following formula:

$$\widehat{W} = \widehat{\lambda}^{1/\varepsilon}$$

- Where λ is the share of expenditure coming from domestic goods
- ε is the elasticity of trade
- Formula only good for *ex-post* evaluation – unless stronger assumptions are imposed
- But can also give us the gains comparing to autarky.

The Armington Model

NOTE: Reshuffle of notations, I try to follow their paper as close as I can.

- Goods differentiated by country of origin
- One unit of final good produced with one unit of labor
- $U_j = \left[\sum_{i=1}^{n} q_{ij}^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$ and $P_j = \left[\sum_{i=1}^{n} \left(w_i \tau_{ij}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$
- $X_{ij} = \left(\frac{w_i \tau_{ij}}{P_j}\right)^{1-\sigma} Y_j$, where $Y_j \equiv \sum_{i=1}^{n} X_{ij}$ is total expenditure in country j (note: X_{ij} now denotes exports from i to j)
- $\frac{\partial \ln\left(\frac{X_{ij}}{X_{jj}}\right)}{\partial \ln \tau_{ij}} = 1 - \sigma < 0$ is the partial elasticity of relative imports w.r.t. variable trade costs.

Foreign Shock

- Fix a given country j, consider foreign shock that affects labor endowments $\mathbf{L} \equiv \{L_i\}$, and trade costs, $\tau \equiv \{\tau_{ij}\}$
- But leaves L_j and τ_{jj} unchanged
- What is the change in real income $\frac{Y_j}{P_j}$ caused by the shock?
- Numeraire: labor in country $j \Rightarrow w_j = 1$
- Trade balance implies $Y_j = w_j L_j \Rightarrow d\ln Y_j = d\ln w_j \Rightarrow d\ln Y_j = 0$
- $d\ln W_j = -d\ln P_j$

Some Algebra

- $P_j = \left[\sum_{i=1}^{n} \left(e^{\ln w_i} e^{\ln \tau_{ij}} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$
- Consider infinitesimal change in trade costs τ
- $d \ln P_j = \sum_{i=1}^{n} \left(\frac{\partial \ln P_j}{\partial \ln w_i} d \ln w_i + \frac{\partial \ln P_j}{\partial \ln \tau_{ij}} d \ln \tau_{ij} \right)$
- $\frac{\partial \ln P_j}{\partial \ln w_i} = \frac{1}{1-\sigma} \frac{(1-\sigma)(w_i \tau_{ij})^{1-\sigma}}{\sum_{i=1}^{n} (w_i \tau_{ij})^{1-\sigma}} = \frac{X_{ij}}{Y_j} = \lambda_{ij}$, and $\frac{\partial \ln P_j}{\partial \ln \tau_{ij}} = \lambda_{ij}$
- $d \ln P_j = \sum_{i=1}^{n} \lambda_{ij} \left(d \ln w_i + d \ln \tau_{ij} \right)$
- Change in price index is a weighted average of price changes all over the world
- Key idea throughout paper: *We can express price changes as a function of changes in expenditure shares.*

Welfare Change

- $X_{ij} = \left(\frac{w_i \tau_{ij}}{P_j}\right)^{1-\sigma} Y_j \Rightarrow \frac{\lambda_{ij}}{\lambda_{jj}} = \left(\frac{w_i \tau_{ij}}{w_j \tau_{jj}}\right)^{1-\sigma}$
- $d \ln \lambda_{ij} - d \ln \lambda_{jj} = (1-\sigma)\left(d \ln w_i + d \ln \tau_{ij}\right)$
- $d \ln W_j = \frac{\sum_{i=1}^{n} \lambda_{ij}\left(d \ln \lambda_{jj} - d \ln \lambda_{ij}\right)}{1-\sigma} = \frac{d \ln \lambda_{jj}}{1-\sigma}$
  - $\sum_{i=1}^{n} \lambda_{ij} = 1 \Rightarrow \sum_{i=1}^{n} \lambda_{ij} d \ln \lambda_{ij} = 0$
- $\widehat{W}_j = \widehat{\lambda}_{jj}^{1/\varepsilon}$, here $\varepsilon = 1 - \sigma$
- Important: in Armington model there are no production gains from trade. No reallocation, no productivity effects. Welfare effects depend only on the elasticity of substitution σ.

Setting

- $i = 1, ..., n$ countries
- 1 factor of production: labor
- Multiple goods $\omega \in \Omega$
- $\overline{N}$: measure of goods in Ω
- L_i : labor endowment in country i
- w_i : wage in country i

Preferences

- Dixit-Stiglitz preferences
- Elasticity of substitution $\sigma > 1$
- $P_i = \left[\int_{\omega \in \Omega} p_i(\omega)^{1-\sigma} \, d\omega\right]^{\frac{1}{1-\sigma}}$
- $p_i(\omega) = +\infty$ if good ω is not available in country i

Technology

- For each $\omega \in \Omega$: a blueprint can be acquired by one or more firms (depends on market structure, i.e. EK vs. Krugman/Melitz)
- Firm in country i produces quantities $\mathbf{q} \equiv \{q_j\}$ of good ω
- Cost function

$$C_i(\mathbf{w}, \mathbf{q}, \omega) = \sum_{j=1}^{n} \left(c_{ij}(w_i, \omega) q_j + f_{ij}(w_i, w_j, \omega) 1(q_j > 0) \right)$$

- $c_{ij}(w_i, \omega)$: constant marginal cost
- $f_{ij}(w_i, w_j, \omega)$: fixed cost of exporting, may depend on both w_i and w_j

Technology

- $c_{ij}(w_i, \omega) = \tau_{ij} w_i \alpha_{ij}(\omega)$
- $f_{ij}(w_i, w_j, \omega) = \xi_{ij} h_{ij}(w_i, w_j) \phi_{ij}(\omega)$

- $h_{ij}(w_i, w_j) > 0$ homogeneous of degree one - captures importance of domestic and foreign labor in fixed costs of exporting
- $\alpha_{ij}(\omega) \geq 0$, $\phi_{ij}(\omega)$ exogenous heterogeneous costs across blueprints

Market Structure

- Perfect Competition

- Monopolistic Competition
  - Restricted Entry
  - Free Entry

- In both structures:
  - Large number of firms
  - Markets clear

Perfect Competition

- Firms have free access to all blueprints
- No fixed costs of exporting: $\phi_{ij}(\omega) = 0 \; \forall i, j, \omega$
- Firms maximize profits taking wages and prices as given

Monopolistic Competition

- Restricted Entry
  - Exogenous measure of firms $\overline{N}_i < \overline{N}$ receive monopoly power over a blueprint
- Free Entry
  - Firms from country i need to hire $F_i > 0$ units of labor in order to acquire monopoly power over one blueprint.
  - Assignment of blueprints to firms is random
  - Measure of goods N_i obtained endogenously
  - Entry costs $w_i F_i$ = expected profits
- Firms maximize profits taking wages, total expenditure and consumer prices as given

Macro-Level Restrictions

- Restrict attention to models satisfying 3 macro-level restrictions

1. Trade is balanced
2. Aggregate Profits (sum across destinations) are a constant share of GDP
3. Import demand system is CES

Trade is balanced

- $X_{ij} =$ Country j's imports from country i
- $X_{ij} = \int_{\omega \in \Omega} x_{ij}(\omega)\, d\omega$
- Value of imports = Value of exports

$$\textbf{R1}: \sum_{i=1}^{n} X_{ij} = \sum_{i=1}^{n} X_{ji}$$

- Total Expenditure in country j: $Y_j \equiv \sum_{i=1}^{n} X_{ij}$ is equal to country j's total revenues $R_j = \sum_{i=1}^{n} X_{ji}$

Aggregate Profits

- Aggregate Profits are a constant share of revenues
- **R2** : $\frac{\Pi_j}{R_j}$ is a constant across j
- This restriction holds trivially under perfect competition ($\Pi_j = 0 \; \forall j$)

Import Demand System is CES

- Import demand is a mapping from $(\mathbf{w}, \mathbf{N}, \tau)$ into $\mathbf{X} \equiv \{X_{ij}\}$
  - $\mathbf{w} \equiv \{w_i\}$ vector of wages
  - $\mathbf{N} \equiv \{N_i\}$ vector of measures of goods produced in each country
  - $\tau \equiv \{\tau_{ij}\}$ matrix of variable trade costs
- The mapping is determined by
  - Profit maximization given preferences
  - Technological constraints
  - Market structure
  - **It excludes labor market clearing and free entry conditions determining the equilibrium values of $\mathbf{w}$ and $\mathbf{N}$.**

Import Demand System is CES

- **R3** $\varepsilon_j^{ii'} = \frac{\partial \ln\left(\frac{X_{ij}}{X_{jj}}\right)}{\partial \ln\left(\tau_{i'j}\right)} = \begin{cases} \varepsilon < 0 \text{ if } i = i' \\ \text{(important: no heterogeneity)} \\ 0 \text{ otherwise} \end{cases}$
- Consider Mexico, US and Canada
  - A reduction in trade costs between Mexico and the US, has no effect on $\frac{X_{US,Canada}}{X_{Canada,Canada}}$
  - It has the same effect on $X_{US,Canada}$ as in $X_{Canada,Canada}$
- Stronger version
- **R3'** $X_{ij} = \frac{\chi_{ij} N_i (w_i \tau_{ij})^{\varepsilon} Y_j}{\sum_{i'=1}^{n} \chi_{i'j} N_{i'} \left(w_{i'} \tau_{i'j}\right)^{\varepsilon}}$
- With this stronger version, we can conduct **ex-ante** policy analysis and compute $\widehat{\lambda}_{jj}$ **before** observing the trade shock – see paper for details.

Welfare Evaluation

- Definition: A foreign shock in country j is a change from $(\mathbf{L}, \mathbf{F}, \tau, \xi)$ to $\left(\mathbf{L}', \mathbf{F}', \tau', \xi'\right)$ such that

$$L_j = L_j' \, ; F_j = F_j' \, ; \tau_{jj} = \tau_{jj}' \, ; \xi_{jj} = \xi_{jj}'$$

- Proposition 1
- Suppose that restrictions R1-R3 hold. Then the change in real income associated with any foreign shock in country j can be computed as

$$\widehat{W}_j = \widehat{\lambda}_{jj}^{1/\varepsilon}$$

- Ex post welfare evaluation

Welfare Evaluation

- Ex ante welfare evaluation: move to autarky ($\lambda_{jj}^{A} = 1$)

$$\widehat{W}_{j}^{A} = \lambda_{jj}^{-1/\varepsilon}$$

Summary of Features of the Models Considered

- Four Primitive Assumptions
  - Dixit-Stiglitz preferences
  - One factor of production (labor)
  - Linear cost functions
  - Perfect or monopolistic competition
- + 3 macro-level restrictions
  - R1 : trade is balanced
  - R2 : aggregate profits are a constant share of aggregate revenues
  - R3 : the aggregate demand system is CES

Example 1: Ricardian Model

- $c_{ij}\left(w_i, \omega\right) = \tau_{ij} w_i \alpha_i\left(\omega\right)$
- $f_{ij}\left(w_i, w_j, \omega\right) = 0$

- Perfect Competition
- $\alpha_{ij}\left(\omega\right) = \alpha_i\left(\omega\right)$
- $\phi_{ij}\left(\omega\right) = 0$
- Under these assumptions, the general trade model becomes a Ricardian model of trade.

Example 1: Ricardian Model

- Let $G(\alpha_1, ..., \alpha_n)$ the prob for any good $\omega \in \Omega$ such that $\alpha_i(\omega) \leq \alpha_i \ \forall i$
- $g(\alpha_1, ..., \alpha_n)$ is the density of G
- Under Perfect Competition, the set of goods that country j buys from country is given by

$$\Omega_{ij} = \left\{ \omega \in \Omega | c_{ij}\alpha_{ij}(\omega) < c_{i'j}\alpha_{i'j}(\omega) \ \forall i' \neq i \right\}$$
$$c_{ij} = w_i \tau_{ij}$$

- Dixit-Stiglitz preferences imply:

$$x_{ij}(\omega) = Y_j \left(\frac{p_{ij}(\omega)}{P_j} \right)^{1-\sigma}$$
$$P_j = \left(\sum_{i'=1}^{n} \int_{\omega \in \Omega_{i'j}} p_{i'j}(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}$$

Aggregate Trade Flows

- $X_{ij} = \int_{\omega \in \Omega_{ij}} x_{ij}(\omega)\, d\omega = Y_j \frac{\int_{\omega \in \Omega_{ij}} p_{ij}(\omega)^{1-\sigma} d\omega}{\sum_{i'=1}^{n} \int_{\omega \in \Omega_{i'j}} p_{i'j}(\omega)^{1-\sigma} d\omega}$

- $X_{ij} = \frac{\int_0^{+\infty} (c_{ij}\alpha_i)^{1-\sigma} g_i(\alpha_i, c_{1j}, \ldots, c_{nj}) d\alpha_i}{\sum_{i'=1}^{n} \int_0^{+\infty} \left(c_{i'j}\alpha_i\right)^{1-\sigma} g_{i'}(\alpha_{i'}, c_{1j}, \ldots, c_{nj}) d\alpha_i} Y_j$

- $g_i(\alpha_i, c_{1j}, \ldots, c_{nj}) = \int_{\alpha_1 > \frac{\alpha_i c_{ij}}{c_{1j}}} \ldots \int_{\alpha_{i-1} > \frac{\alpha_i c_{ij}}{c_{i-1j}}} \int_{\alpha_{i+1} > \frac{\alpha_i c_{ij}}{c_{i+1j}}} \ldots \int_{\alpha_n > \frac{\alpha_i c_{ij}}{c_{nj}}} g(\alpha)\, d\alpha_{-i}$ i.e. the conditional density of α_i given it's the lowest cost producer

Trade Elasticities

- $\frac{\partial \ln\left(\frac{X_{ij}}{X_{jj}}\right)}{\partial \ln \tau_{i'j}} = \varepsilon_j^{ii'} = \begin{cases} 1-\sigma+\gamma_{ij}^i-\gamma_{jj}^i \text{ for } i' = i \\ \gamma_{ij}^{i'}-\gamma_{jj}^{i'} \text{ for } i' \neq i \end{cases}$

- $1-\sigma$ is an intensive margin elasticity

- $\gamma_{ij}^{i'} \equiv \frac{\partial \ln\left[\int_0^{+\infty} \alpha_i^{1-\sigma} g_i(\alpha_i, c_{1j}, \ldots, c_{nj}) d\alpha_i\right]}{\partial \ln c_{i'j}}$ is an extensive margin elasticity
  - if τ_{ij} is reduced, goods previously not exported from i to j will now be exported
  - if τ_{ij} is reduced, goods previously produced and consumed in j will now be imported from i

Welfare Predictions

- Aggregate Budget Constraint implies: $Y_j = w_j L_j$
- $d \ln Y_j = d \ln w_j + d \ln L_j = d \ln w_j = 0$ (labor at j is the numeraire)
- $d \ln W_j = -d \ln P_j = -\sum_{i=1}^{n} \frac{\partial \ln P_j}{\partial \ln c_{ij}} d \ln c_{ij} = -\sum_{i=1}^{n} \lambda_{ij} d \ln c_{ij}$

Welfare Predictions

- $X_{ij} = \frac{\int_0^{+\infty} (c_{ij}\alpha_i)^{1-\sigma} g_i(\alpha_i, c_{1j}, ..., c_{nj}) d\alpha_i}{\sum_{i'=1}^{n} \int_0^{+\infty} \left(c_{i'j}\alpha_i\right)^{1-\sigma} g_{i'}(\alpha_{i'}, c_{1j}, ..., c_{nj}) d\alpha_i} Y_j$

- $\ln \lambda_{ij} - \ln \lambda_{jj} =$
  $\ln \int_0^{+\infty} (c_{ij}\alpha_i)^{1-\sigma} g_i(\alpha_i, c_{1j}, ..., c_{nj}) d\alpha_i - \ln \int_0^{+\infty} (c_{jj}\alpha_j)^{1-\sigma} g_j(\alpha_j, c_{1j}, ..., c_{nj}) d\alpha_j$

- $d \ln \lambda_{ij} - d \ln \lambda_{jj} = \left(1 - \sigma + \gamma_{ij}^i - \gamma_{jj}^i\right) d \ln c_{ij} + \sum_{i' \neq i,j}^{n} \left(\gamma_{ij}^{i'} - \gamma_{jj}^{i'}\right) d \ln c_{i'j}$

- $d \ln W_j = -\sum_{i=1}^{n} \lambda_{ij} \left[\frac{d \ln \lambda_{ij} - d \ln \lambda_{jj}}{1 - \sigma + \gamma_{ij}^i - \gamma_{jj}^i} - \sum_{i' \neq i,j}^{n} \left(\frac{\gamma_{ij}^{i'} - \gamma_{jj}^{i'}}{1 - \sigma + \gamma_{ij}^i - \gamma_{jj}^i} \right) d \ln c_{i'j} \right]$

Welfare Predictions

- To derive Proposition 1, note that R3 implies that
- $\gamma_{ij}^{i'} - \gamma_{jj}^{i'} = 0$ for all $i' \neq i, j$ and that $1 - \sigma + \gamma_{ij}^{i} - \gamma_{jj}^{i} = \varepsilon$ for all $i \neq j$
- $$d \ln W_j = -\sum_{i=1}^{n} \lambda_{ij} \left[\frac{d \ln \lambda_{ij} - d \ln \lambda_{jj}}{1 - \sigma + \gamma_{ij}^{i} - \gamma_{jj}^{i}} - \sum_{i' \neq i,j}^{n} \left(\frac{\gamma_{ij}^{i'} - \gamma_{jj}^{i'}}{1 - \sigma + \gamma_{ij}^{i} - \gamma_{jj}^{i}} \right) d \ln c_{i'j} \right] \Rightarrow$$
  $$d \ln W_j = -\sum_{i=1}^{n} \lambda_{ij} \left[\frac{d \ln \lambda_{ij} - d \ln \lambda_{jj}}{\varepsilon} \right]$$
- $$\sum_{i=1}^{n} \lambda_{ij} = 1 \Rightarrow \sum_{i=1}^{n} \lambda_{ij} d \ln \lambda_{ij} = 0 \Rightarrow d \ln W_j = \frac{d \ln \lambda_{jj}}{\varepsilon}$$
- $$\widehat{W}_j = \widehat{\lambda}_{jj}^{1/\varepsilon}$$

Role of Macro-level Restrictions

- R1 and R2 are trivially satisfied under perfect competition
- R3 is the crucial restriction that guarantees that:

1. Changes in each component of the price index $d \ln c_{ij}$ can be recovered one to one from changes in relative imports $d \ln \lambda_{ij} - d \ln \lambda_{jj}$ (because $\gamma_{ij}^{i'} - \gamma_{jj}^{i'} = 0$ for all $i' \neq i, j$)
2. Changes in relative imports can be aggregated into changes in the domestic share of expenditure $-d \ln \lambda_{jj}$ (because $1 - \sigma + \gamma_{ij}^{i} - \gamma_{jj}^{i} = \varepsilon$ for all $i \neq j$)
3. Small changes in the domestic expenditures can be integrated into larger ones (because the trade elasticity ε is constant).

When are the Macro-level Restrictions satisfied?

- $\frac{\partial \ln\left(\frac{X_{ij}}{X_{jj}}\right)}{\partial \tau_{i'j}} = \varepsilon_j^{ii'} = \begin{cases} 1-\sigma+\gamma_{ij}^i-\gamma_{jj}^i \text{ for } i^{'}=i \\ \gamma_{ij}^{i'}-\gamma_{jj}^{i'} \text{ for } i^{'}\neq i \end{cases}$
- In general R3 is not going to be satisfied
- Eaton and Kortum (2002) provides an example of a Ricardian model satisfying R3
- That model is obtained assuming that $\frac{1}{\alpha_i(\omega)}$ are independently drawn from Frechet distributions
- $g\left(\alpha_1,...,\alpha_n\right)=\prod\theta T_i\alpha_i^{\theta-1}e^{-T_i\alpha_i^{\theta}}$ for all $\alpha_1,...,\alpha_n\geq 0$
- Tedious algebra yields

$$X_{ij}=\frac{T_i\left(w_i\tau_{ij}\right)^{-\theta}}{\sum_{i'=1}^{n}T_{i'}\left(w_{i'}\tau_{i'j}\right)^{-\theta}}Y_j$$

- Which satisfies R3', a stronger form of R3

Eaton and Kortum vs Armington

- EK has more margins of adjustment than Armington – intensive + extensive margins operate.
- But same formula for welfare gains?
- They have the same formulas, but ε is different!
- In EK, $\varepsilon = \theta > \sigma - 1$, so larger gains in EK

Example 2: Melitz (2003)

- Monopolistic competition with free entry
- Firm heterogeneity
- good-specific unit labor requirements do not vary across destinations, $\alpha_{ij}(\omega) = \alpha_i(\omega)$
- No heterogeneity in fixed costs $\phi_{ij}(\omega) = 1$ for all i, j and ω
- $G(\alpha_1, ..., \alpha_n)$: share of goods $\omega \in \Omega$ with $\alpha_i(\omega) \leq \alpha_i$ for all i
- density function : $g(\alpha_1, ..., \alpha_n)$
- $h_{ij}(w_i, w_j) = w_i^{\mu} w_j^{1-\mu}$ with $0 \leq \mu \leq 1$

Aggregate Trade Flows

- Firms charge constant markup $\frac{\sigma}{\sigma-1}$ over MC
- Profits
  - $\pi_{ij}(\omega) = \left[\frac{\sigma c_{ij}\alpha_{ij}(\omega)}{(\sigma-1)P_j}\right]^{1-\sigma}\left(\frac{Y_j}{\sigma}\right) - \xi_{ij}w_i^{\mu}w_j^{1-\mu}$
  - $c_{ij} \equiv w_i\tau_{ij}$
- Price index

$$P_j = \left(\sum_{i=1}^{n}\int_{\omega\in\Omega_{ij}}\frac{\sigma}{\sigma-1}\left(c_{ij}\alpha_i(\omega)\right)^{1-\sigma}d\omega\right)^{\frac{1}{1-\sigma}}$$

Aggregate Trade Flows

- α_{ij}^* : cutoff determining entry of firms from country i in country j
  - $\pi_{ij}(\omega) > 0 \Leftrightarrow \alpha_{ij}(\omega) < \alpha_{ij}^*$
- $\Omega_{ij} = \left\{ \omega \in \Omega | \alpha_{ij}(\omega) < \alpha_{ij}^* = \sigma^{\frac{\sigma}{1-\sigma}} (\sigma - 1) \left(\frac{P_j}{c_{ij}} \right) \left(\frac{\xi_{ij} w_i^{\mu} w_j^{1-\mu}}{Y_j} \right)^{\frac{1}{1-\sigma}} \right\}$
- Price index

$$P_j = \left(\sum_{i=1}^{n} \int_0^{\alpha_{ij}^*} \frac{\sigma}{\sigma - 1} (c_{ij} \alpha_i)^{1-\sigma} N_i g_i(\alpha_i) \, d\alpha_i \right)^{\frac{1}{1-\sigma}}$$

- $X_{ij} = \frac{N_i \int_0^{\alpha_{ij}^*} (c_{ij} \alpha_i)^{1-\sigma} g_i(\alpha_i) d\alpha_i}{\sum_{i'=1}^{n} N_{i'} \int_0^{\alpha_{i'j}^*} \left(c_{i'j} \alpha_{i'} \right)^{1-\sigma} g_{i'}(\alpha_{i'}) d\alpha_{i'}} Y_j$

Trade Elasticities

- $$\frac{\partial \ln\left(\frac{X_{ij}}{X_{jj}}\right)}{\partial \ln \tau_{i'j}} = \varepsilon_j^{ii'} = \begin{cases} 1 - \sigma - \gamma_{ij} + (\gamma_{ij} - \gamma_{jj}) \left(\frac{\partial \ln \alpha_{jj}^*}{\partial \ln \tau_{ij}}\right) & \text{for } i' = i \\ (\gamma_{ij} - \gamma_{jj}) \left(\frac{\partial \ln \alpha_{jj}^*}{\partial \ln \tau_{ij}}\right) & \text{for } i' \neq i \end{cases}$$
- $\gamma_{ij} \equiv \frac{d \ln \int_0^{\alpha_{ij}^*} \alpha^{1-\sigma} g_i(\alpha) d\alpha}{d \ln \alpha_{ij}^*}$ is an extensive margin elasticity
- It is easier to see the above if we use the relationships $\frac{\partial \ln \alpha_{ij}^*}{\partial \ln \tau_{ij}} = \frac{\partial \ln \alpha_{jj}^*}{\partial \ln \tau_{ij}} - 1$ and $\frac{\partial \ln \alpha_{ij}^*}{\partial \ln \tau_{i'j}} = \frac{\partial \ln \alpha_{jj}^*}{\partial \ln \tau_{i'j}}$

Welfare

- The representative agent's budget constraint still implies $d \ln Y_j = d \ln w_j = 0$, so $d \ln W_j = -d \ln P_j$.
- Start with $(1-\sigma) d \ln P_j = \sum_{i=1}^{n} \lambda_{ij} (1-\sigma) \left(d \ln c_{ij} + d \ln N_i + \gamma_{ij} d \ln \alpha_{ij}^* \right)$
- Use the fact that $d \ln \alpha_{ij}^* = d \ln P_j - d \ln \tau_{ij} + \frac{(d \ln \xi_{ij} + \mu d \ln w_i)}{1-\sigma}$
- We then have

$$[(1-\sigma) - \gamma_j] d \ln P_j = \sum_{i=1}^{n} \lambda_{ij} \left[\begin{array}{c} (1-\sigma-\gamma_{ij}) \, d \ln \tau_{ij} + \left(1 - \sigma + \gamma_{ij} \frac{\mu}{1-\sigma} \right) d \ln w_i \\ \frac{\gamma_{ij}}{1-\sigma} d \ln \xi_{ij} + d \ln N_i \end{array} \right], \text{ where}$$

$$\gamma_j = \sum_{i=1}^{n} \lambda_{ij} \gamma_{ij}$$

- As a result $d \ln P_j = \sum_{i=1}^{n} \frac{\lambda_{ij}}{[(1-\sigma)-\gamma_j]} \left[\begin{array}{c} (1-\sigma-\gamma_{ij}) \, d \ln \tau_{ij} + \left(1 - \sigma + \gamma_{ij} \frac{\mu}{1-\sigma} \right) d \ln w_i \\ \frac{\gamma_{ij}}{1-\sigma} d \ln \xi_{ij} + d \ln N_i \end{array} \right]$

Welfare

- Trade flow implies that
  $d \ln \lambda_{ij} - d \ln \lambda_{jj} = (1-\sigma)(d \ln w_i + d \ln \tau_{ij}) + \gamma_{ij} d \ln \alpha^*_{ij} - \gamma_{jj} d \ln \alpha^*_{jj} + d \ln N_i - d \ln N_j$
- We use the definition $d \ln \alpha^*_{ij} = d \ln \alpha^*_{jj} - d \ln \tau_{ij} + \frac{\mu}{1-\sigma} d \ln w_i + \frac{d \ln \xi_{ij}}{1-\sigma}$
- Substitute the above trade flow into Price index
  $d \ln P_j = \sum_{i=1}^{n} \left(\frac{\lambda_{ij}}{1-\sigma-\gamma_j} \right) \left[d \ln \lambda_{ij} - d \ln \lambda_{jj} - (\gamma_{ij} - \gamma_{jj}) \, d \ln \alpha^*_{jj} + d \ln N_j \right]$
- R3 needs $\gamma_{ij} = \gamma_{jj}$ and $1 - \sigma - \gamma_j = \varepsilon$ for all i, j
- $d \ln W_j = -d \ln P_j = \frac{d \ln \lambda_{jj} - d \ln N_j}{\varepsilon}$, which needs $\Pi_j = N_j F_j$ (Free Entry) and proportional to Y_j, by R1 and R2, $\Rightarrow d \ln N_j = d \ln Y_j = 0$
- $\widehat{W}_j = \widehat{\lambda}_{jj}^{1/\varepsilon}$
- Now R2 plays a crucial role. **For R3 to be satisfied: Pareto distribution of productivity** ($\gamma_{ij} = \gamma_{jj} = 1 - \sigma + \theta$).

Outline

1 Gravity: Facts

2 Micro-foundations

3 Structural Gravity

4 Gains from Trade (ACR)

5 Taking ACR to the Data

Estimating the Trade Elasticity

- Gravity Equation

$$\ln X_{ij} = A_i + B_j + \varepsilon \ln \tau_{ij} + \nu_{ij}$$

- Conditional on a trade elasticity ε and on the change in fraction of domestic expenditure $\widehat{\lambda}_{jj}$ we can compute gains from trade within a relatively large class of models of trade. You don't need to pick exactly what model you have in mind.

- Problem: τ_{ij} is not observed. Identification requires an observable component of trade costs whose elasticity is known:
  - Tariffs: $\tau_{ij} = (1 + t_{ij})\, d_{ij} \Rightarrow$ coefficient on $\ln(1 + t_{ij})$ is ε (Caliendo and Parro, 2015, use triple differences across countries)
  - Price gaps: EK (2002) use the (near-)maximum retail price gap across 50 goods as a proxy for $\ln \tau_{ij}$; Simonovska and Waugh (2014) correct the small-sample bias of the max operator
  - Distance: only identifies $\varepsilon\rho$, not ε separately

How Large Is the Trade Elasticity?

Study	Source of variation	$\lvert\varepsilon\rvert$
Eaton and Kortum (2002)	price gaps, wages	8.28 (baseline)
Simonovska and Waugh (2014)	price gaps, bias-corrected	≈ 4
Head and Mayer (2014)	meta-analysis of tariff coef.	median ≈ 5
Caliendo and Parro (2015)	sectoral tariffs	varies by sector
Boehm, Levchenko and Pandalai-Nayar (2023)	tariff changes, local projections	0.76 (1 year) 1.75–2.25 (long run)

- Estimates from tariff *changes* are much smaller than those from cross-sectional variation; the elasticity rises with the horizon
- Interpretation for ACR: ε should be the long-run elasticity relevant for the comparative static in question
- Gains from trade scale with $1/|\varepsilon|$: the choice of ε matters more than the choice of model

Gains from Trade: Back-of-the-Envelope

- Gains from trade relative to autarky: $GT_j = 1 - \lambda_{jj}^{-1/\varepsilon}$ (share of real income lost in autarky)

	$\varepsilon = -2$	$\varepsilon = -4$	$\varepsilon = -5$	$\varepsilon = -8.28$
$\lambda_{jj} = 0.91$	4.6%	2.3%	1.9%	1.1%
$\lambda_{jj} = 0.75$	13.4%	6.9%	5.6%	3.4%
$\lambda_{jj} = 0.60$	22.5%	12.0%	9.7%	6.0%

- Costinot and Rodríguez-Clare (2014): one-sector model with $\varepsilon = -5$ gives 1.8% for the U.S.
- Adding multiple sectors and input-output linkages raises gains substantially (8% for the U.S. in their benchmark with intermediates; cross-country average around 27–32%)
  - Sectors with low $|\varepsilon_s|$ and low $\lambda_{jj,s}$ contribute disproportionately (Ossa, 2015)
  - Intermediates amplify: cheaper imported inputs lower the cost of all domestic production

Ex-Ante Counterfactuals: Exact Hat Algebra

- Dekle, Eaton and Kortum (2007, 2008): solve for counterfactual changes $\hat{x} = x'/x$ using only baseline data and ε
- Armington/EK with labor as the only factor, $Y_i = w_i L_i$, trade deficit D_j, shock $\{\hat{\tau}_{ij}\}$:

$$\lambda'_{ij} = \frac{\lambda_{ij}\left(\hat{w}_i\hat{\tau}_{ij}\right)^{\varepsilon}}{\sum_k \lambda_{kj}\left(\hat{w}_k\hat{\tau}_{kj}\right)^{\varepsilon}} \qquad \text{(gravity)}$$

$$\hat{w}_i Y_i = \sum_j \lambda'_{ij}\left(\hat{w}_j Y_j + D'_j\right) \qquad \text{(market clearing)}$$

- Unknowns: N wage changes $\hat{w}_i$ (one normalization). Data: λ_{ij}, Y_i, D_i. Parameter: ε
- Not needed: technologies T_i, labor L_i, levels of trade costs τ_{ij}; they are all "differenced out"
- Welfare: $\hat{P}_j = \left[\sum_k \lambda_{kj}(\hat{w}_k\hat{\tau}_{kj})^{\varepsilon}\right]^{1/\varepsilon}$ and, with balanced trade, $\widehat{W}_j = \hat{w}_j/\hat{P}_j = \widehat{\lambda}_{jj}^{1/\varepsilon} \Rightarrow$ ACR's ex-ante version (R3')

Exact Hat Algebra: Algorithm

1. Build a balanced dataset: bilateral flows X_{ij} including domestic sales X_{jj} (gross output minus exports), shares $\lambda_{ij} = X_{ij} / \sum_k X_{kj}$, incomes Y_i, deficits D_i
2. Guess $\hat{w}$; normalize $\sum_i \hat{w}_i Y_i = \sum_i Y_i$
3. Compute λ'_{ij} from the gravity equation
4. Compute excess demand $Z_i(\hat{w}) = \frac{1}{Y_i}\left[\sum_j \lambda'_{ij}(\hat{w}_j Y_j + D'_j) - \hat{w}_i Y_i\right]$
5. Update $\hat{w}_i \leftarrow \hat{w}_i\,(1 + \kappa Z_i)$ with a small step $\kappa > 0$; repeat 3–5 until $\max_i |Z_i| <$ tolerance
6. Report $\hat{\lambda}_{jj}$, $\widehat{W}_j$, and changes in trade flows

- Extensions: multiple sectors and input-output linkages (Caliendo and Parro, 2015), tariff revenue (Ossa, 2014), purging deficits first ($D'_j = 0$) as in DEK (2008)
- Same idea underlies "universal gravity" comparative statics (Allen, Arkolakis and Takahashi, 2020)

Extensions

- Multiple sectors and input-output linkages (Costinot and Rodríguez-Clare, 2014; Caliendo and Parro, 2015)
  - $\widehat{W}_j = \prod_s \widehat{\lambda}_{jj,s}^{\eta_{j,s}/\varepsilon_s}$ without intermediates (Cobb-Douglas upper tier with expenditure shares $\eta_{j,s}$); intermediates add a "round-about" multiplier
- Heterogeneous trade elasticities across sectors: Ossa (2015) shows a few sectors with low elasticities account for a large share of the gains
- Non-homothetic preferences: Fajgelbaum and Khandelwal (2016) show that gains from trade are larger for poorer households, who spend more on traded goods
- Non-CES demand: Adao, Costinot and Donaldson (2017) recover counterfactual predictions from the factor demand system without parametric CES assumptions
- Labor market frictions, unemployment, and multiple factors break the one-to-one mapping between λ_{jj} and welfare

Does Firm Heterogeneity Matter for Welfare?

- ACR: given ε and λ_{jj}, micro structure is irrelevant. Two important qualifications:
- Melitz and Redding (2015), "New Trade Models, New Welfare Implications"
  - With untruncated Pareto, the trade elasticity is a constant and ACR applies
  - With more general productivity distributions (e.g. truncated Pareto, log-normal), the trade elasticity is endogenous and varies with trade costs; λ_{jj} and a single elasticity are no longer sufficient
  - Holding structural parameters fixed, firm selection yields larger gains than a homogeneous-firm model
- Arkolakis, Costinot, Donaldson and Rodríguez-Clare (2019), "The Elusive Pro-Competitive Effects of Trade"
  - Variable markups (beyond CES) + Pareto: gains are given by a modified ACR formula
  - Pro-competitive effects are "elusive": lower domestic markups are offset by incomplete pass-through of foreign cost cuts; gains are weakly *lower* than under CES

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